Wenjia Ge

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Research Interests

Robot perception and navigation in real-world environments, including obstacle avoidance and recognizing human states; 3D reconstruction and point-cloud data for embodied AI; human-in-the-loop robotic systems.

Education

New York University, Tandon School of Engineering *Sept 2024 – Dec 2026 (expected)*
New York, NY
M.S., Electrical Engineering, GPA: 3.926/4.0 | GRE: 327
Relevant coursework: Machine Learning, Interactive Medical Robotics, Digital Signal Processing, Real-Time Embedded Systems, Computing Systems Architecture, Applied Matrix Theory, Probability and Stochastic Processes, Systems Engineering

Shenzhen University, Shenzhen, China *Sept 2020 – July 2024*
B.S., Physics, GPA: 3.65/4.0 | Outstanding Undergraduate Thesis Award

Publications

Ge, W., Wen, P., Li, Z., Luo, Q., Zhao, C., Wang, J., & Yang, M. (2023). Application of the Method of Manufactured Solutions to Verify a Nuclear Reactor Physics Computational Code Using the Method of Characteristics. *Proceedings of the 30th International Conference on Nuclear Engineering (ICONE30)*, Paper 1702. doi:10.1299/jsmeicone.2023.30.1702

Contributor, *Wanderland*: a multi-sensor capture and 3D-reconstruction pipeline for embodied-AI simulation (CVPR 2026 Highlight), NYU AI4CE Lab.

Research Experience

Research Assistant, AI4CE Lab (PI: Prof. Chen Feng) *Apr 2026 – Present*
New York University, New York, NY

- Collect real-world indoor and outdoor spatial data with MetaCam and process it into 3D scene representations and point-cloud datasets for embodied-AI navigation research and world-model training.
- Reconstruct captured scenes as 3D Gaussian Splatting environments; build HPC workflows for data ingestion, point-cloud preprocessing, reconstruction, and quality control, and inspect reconstructions in CloudCompare and MeshLab to find artifacts.
- Serve as student team lead for Wanderland, a multi-sensor capture and 3D-reconstruction pipeline for embodied-AI simulation that produced a CVPR 2026 Highlight publication; train new members on capture protocols and track team progress.

Research Assistant: Surgical Navigation Simulation *Feb – Aug 2026*
New York University (PI: Prof. Rui Li), New York, NY

- Designed and built a virtual simulation environment for a concentric-tube surgical robot, modeling robot mechanics and motion in constrained anatomical environments.
- Built a real-time human-in-the-loop pipeline connecting a haptic input device (3D Systems Touch) through ROS2 to RViz and 3D Slicer/SlicerROS2, enabling manual navigation and visualization within vascular anatomy derived from STL models, centerlines, and landmarks (VMTK).
- Owned hardware and driver integration, real-time visualization, and system validation; resolved instability issues from WSL2/USB passthrough.
- Leading ongoing work on tip tracking, coordinate alignment, and constrained navigation for centerline-based progress measurement.
- Built an Arduino and Python data-acquisition workflow to stream and analyze two tactile sensors in real time; characterized baseline noise, contact response, and cross-talk, and tuned detection thresholds.

Research Intern, Embodied AI and Robotics, AI2 Robotics *Oct 2025 – Jan 2026*
Shenzhen, China

- Validated system modules across the perception–decision–action loop, designing test protocols and performance metrics for seven subsystems: perception, localization and mapping, task planning, motion control, manipulation, communications, and safety.
- Injected message delay, packet loss, and sensor disconnection to measure fault detection and recovery time.
- Documented failure cases and supported cross-team troubleshooting across algorithm, hardware, and systems groups; ran three technical demonstrations for prospective partners.

Undergraduate Research: Nuclear Reactor Physics, Shenzhen University *2020 – 2024*
First-author publication at ICONE30

- Applied the Method of Manufactured Solutions to verify the spatial-discretization accuracy of the Method of Characteristics in reactor physics simulation; carried out numerical modeling, coding, debugging, and verification.

Undergraduate Thesis Research: Sb_2S_3 Thin-Film Solar Cells, Shenzhen University *2024*
Outstanding Undergraduate Thesis Award

- Fabricated thin-film solar cells by chemical bath deposition and characterized them with XRD, SEM, J–V, and EQE; raised power conversion efficiency from 4.43% to 5.43% through surface and interface modification.

Teaching Experience

Teaching Assistant, ECE 6113 / BE 6403 Digital Signal Processing I *Fall 2026*
NYU Tandon School of Engineering, New York, NY

- Hold weekly office hours (about 2 hours per week) and answer students' questions on Digital Signal Processing concepts and course material.
- Write midterm and final exam questions and grade student work.
- Coordinate weekly synchronous sessions, lecture recordings, and course materials on Brightspace.

Graduate Student Mentor *Summer 2026*
NYU Tandon School of Engineering, New York, NY

- Advise incoming graduate students on course selection, academic planning, and campus and career resources.

Other Experience

Hardware Test Engineer Intern, Shenzhen Zhi'ao Technology Co., Ltd. *July – Aug 2023*
Shenzhen, China

- Conducted functional and performance testing on embedded motherboards and industrial PCs; wrote C and Python test scripts for serial, Ethernet, and GPIO interfaces; contributed to a 15% reduction in hardware failure rates.

Independent Projects: Audience Behavior & Content Experiments *Sept 2024 – Present*
Self-directed

- Run content experiments across social platforms with 150,000+ combined followers, using engagement data to test hypotheses about user attention.

Honors and Awards

- Mentor Scholarship, NYU Tandon School of Engineering, Fall 2026
- Violet Internship & Research Award (VISAA), New York University
- Outstanding Undergraduate Thesis Award, Shenzhen University

Skills

Programming: Python, MATLAB, C, C++, Verilog, Linux, Git, HPC workflows
Robotics & 3D: ROS2 (Jazzy), RViz, 3D Slicer, SlicerROS2, VMTK, MetaCam, 3D Gaussian Splatting, point-cloud processing, CloudCompare, MeshLab, 3D Systems Touch (haptic device)
Hardware: STM32 (ADC, DMA, timers, UART, SPI, I2C), Arduino, sensor data acquisition
Languages: Mandarin Chinese (Native), English (Fluent)